Pranav Pillai

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PROFESSIONAL SUMMARY

Results-driven software engineer with strong foundation in distributed systems, high-concurrency backend services, and clean architecture. Experienced in designing fault-tolerant microservices, optimizing database throughput, and shipping production code with rigorous test coverage.

EDUCATION

Bachelor of Technology in Computer Science & Engineering — Vellore Institute of Technology *2022 - 2026*

Cumulative Grade Point Average: 9.30/10.0

WORK EXPERIENCE

Backend Engineering Intern — CRED (Dreamplug Technologies), Bengaluru, India *Jun 2025 - Present*

- Developed responsive internal operations dashboard in React and TypeScript with TanStack Query, accelerating merchant dispute resolution by 40%.
- Optimized database queries and added composite b-tree indexes on partitioned ledger tables, lowering average query execution time from 420ms to 35ms.
- Implemented automated circuit-breaker pattern with Resilience4j across 8 downstream microservices, preventing cascade outages during third-party gateway downtime.

TECHNICAL PROJECTS

Autonomous Code Review Linter & Security Auditor | Technologies: Python, AST, Tree-sitter, Docker, FastAPI

- Engineered static analysis pipeline parsing abstract syntax trees (ASTs) to detect SQL injection, unvalidated deserialization, and memory leaks.
- Built automated GitHub Action integration running checks across 200+ pull requests with an average scan duration under 3.5 seconds.
- Containerized execution sandbox using gVisor and Docker to securely isolate untrusted student code submissions.

Distributed Log-Structured Key-Value Store | Technologies: Go, Raft Consensus, gRPC, LSM-Tree, Docker

- Implemented distributed LSM-tree storage engine in Go with write-ahead logging (WAL), SSTable compaction, and bloom filters.
- Embedded Raft consensus algorithm for distributed multi-node replication, surviving arbitrary node crashes without state divergence.
- Benchmarked performance using custom concurrent load generator, achieving 85,000 read operations/sec across a 5-node cluster.

TECHNICAL SKILLS

Languages: Go, Java, Python, TypeScript, C++, SQL

Frameworks & Systems: Spring Boot, Node.js, FastAPI, React, Next.js, gRPC

Databases & Storage: PostgreSQL, Redis, MongoDB, Apache Kafka, Elasticsearch

Cloud & DevOps: Docker, Kubernetes, AWS (EC2, S3, RDS), Git, CI/CD Pipelines